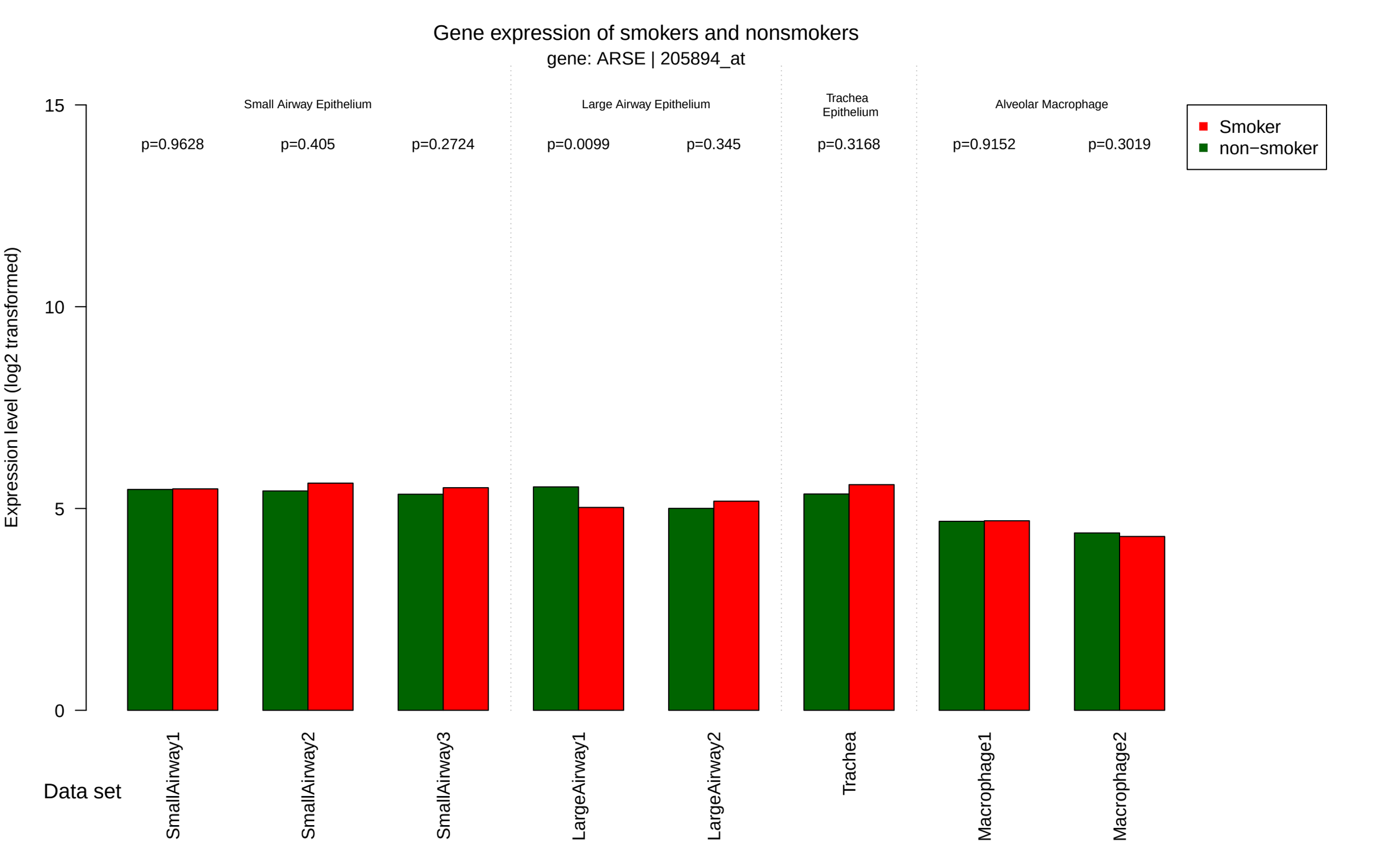
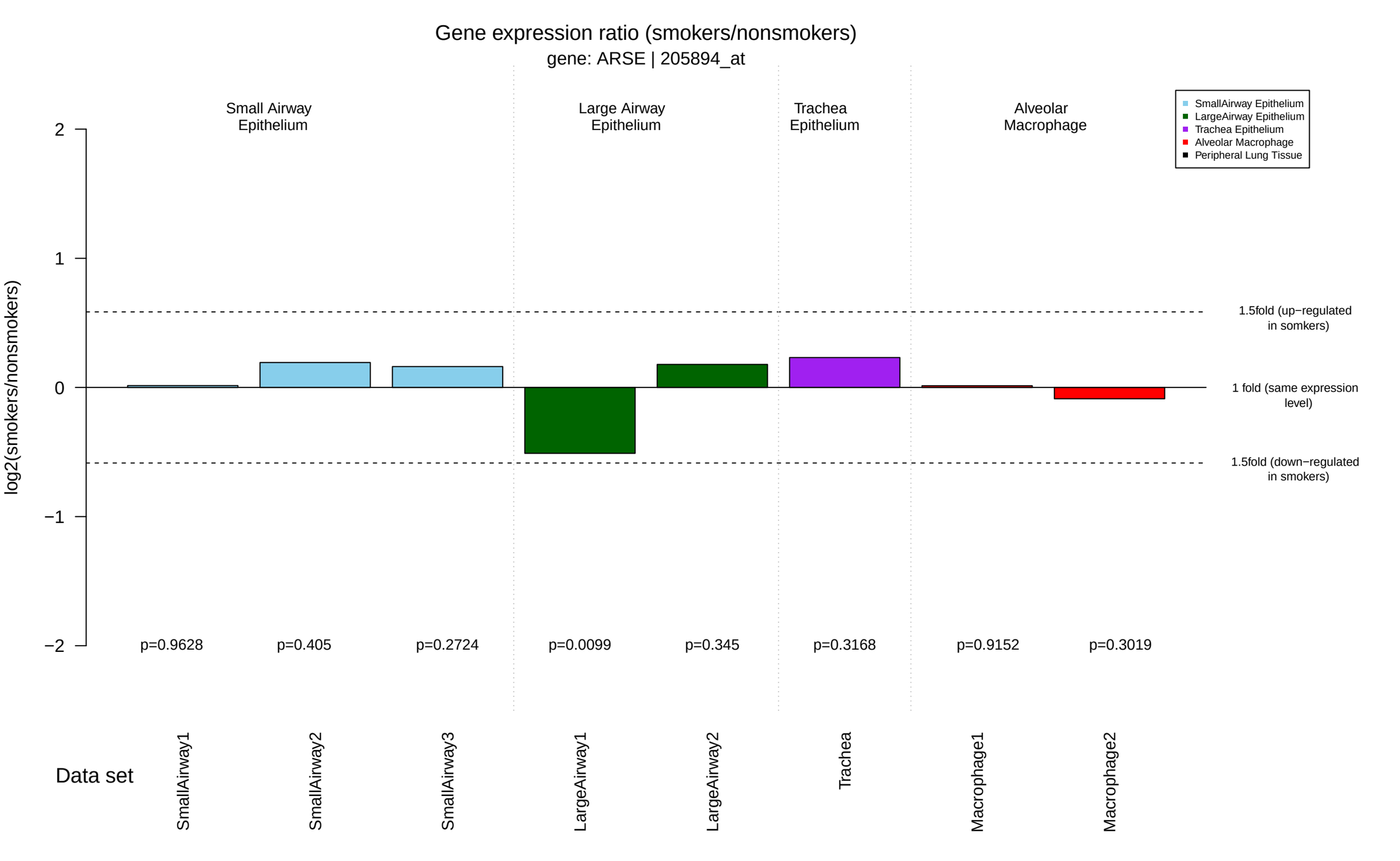
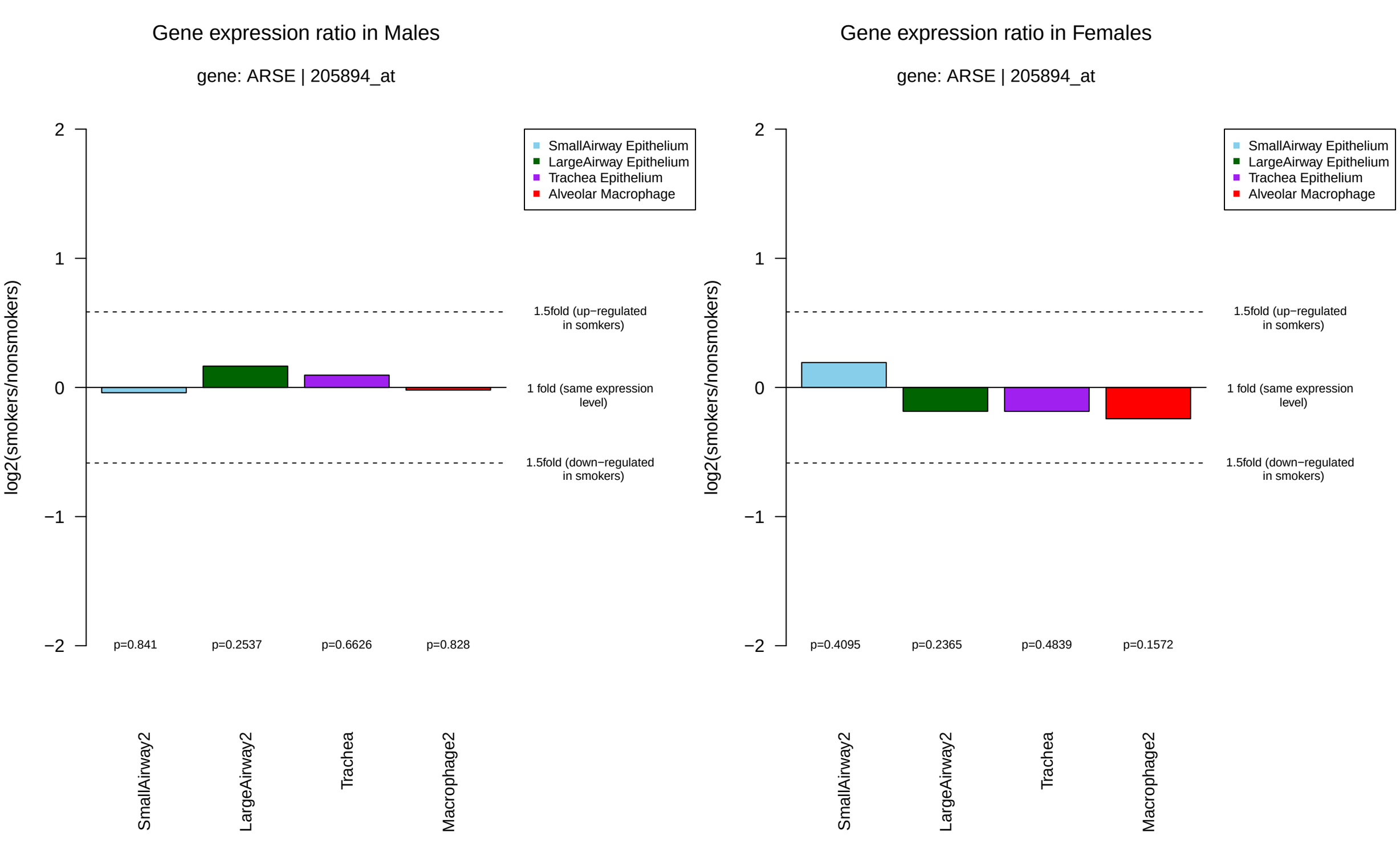
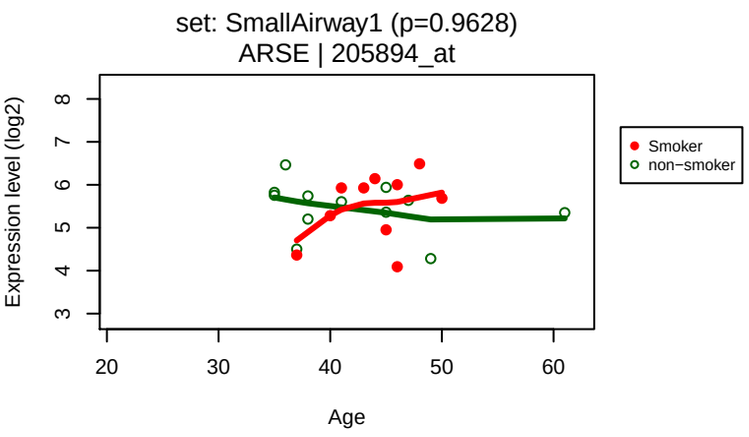
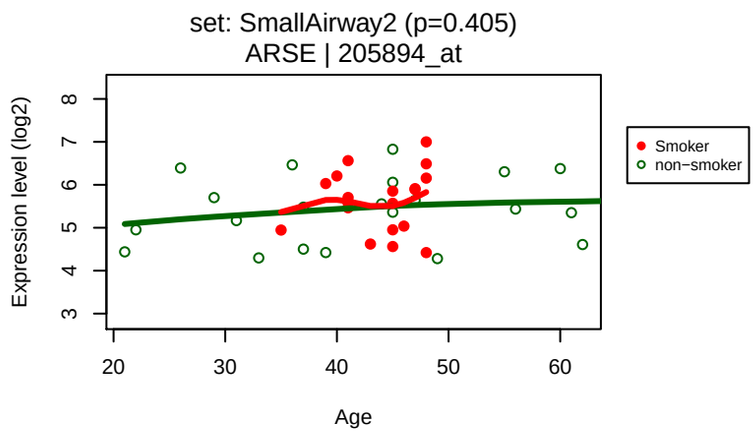
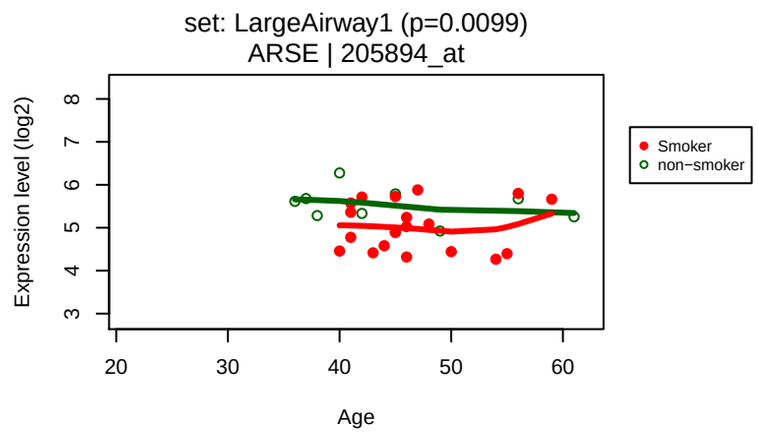

Gene expression of smokers and nonsmokers
gene: ARSE | 205894_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9628
p=0.405
p=0.2724
p=0.0099
p=0.345
p=0.3168
p=0.9152
p=0.3019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

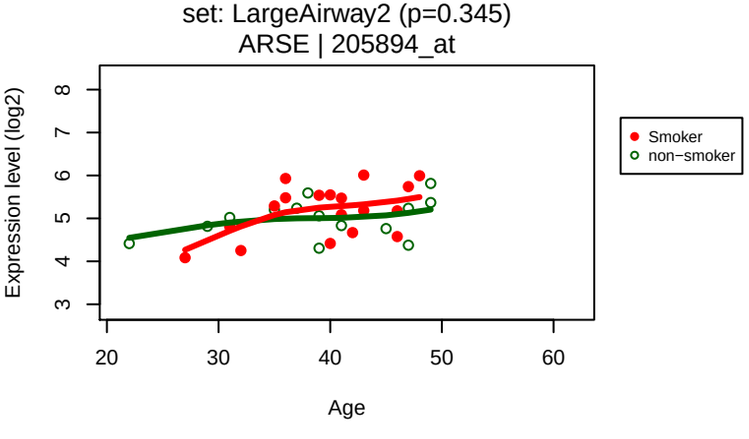

Gene expression ratio (smokers/nonsmokers)
gene: ARSE | 205894_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9628
p=0.405
p=0.2724
p=0.0099
p=0.345
p=0.3168
p=0.9152
p=0.3019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

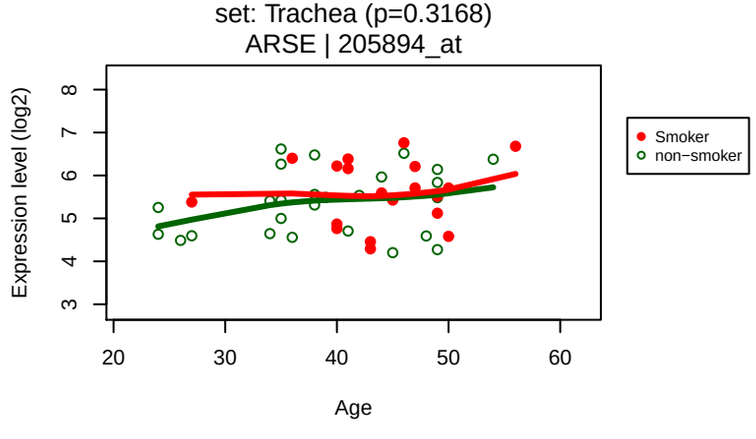

Gene expression ratio in Males
gene: ARSE | 205894_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.841
p=0.2537
p=0.6626
p=0.828
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARSE | 205894_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4095
p=0.2365
p=0.4839
p=0.1572
SmallAirway2
LargeAirway2
Trachea
Macrophage2

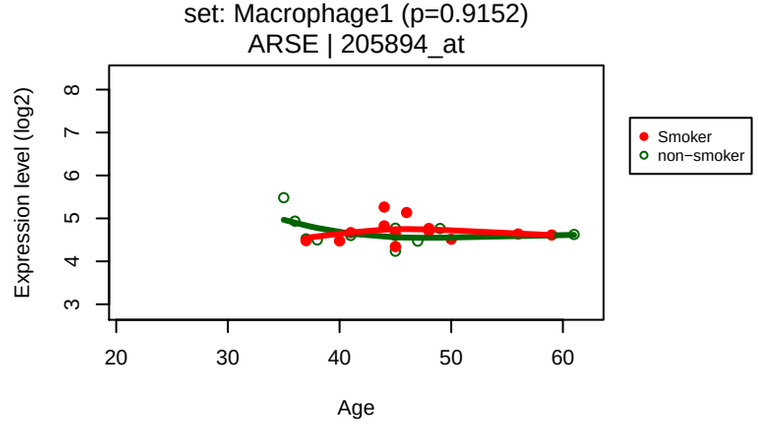

set: SmallAirway1 (p=0.9628)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker

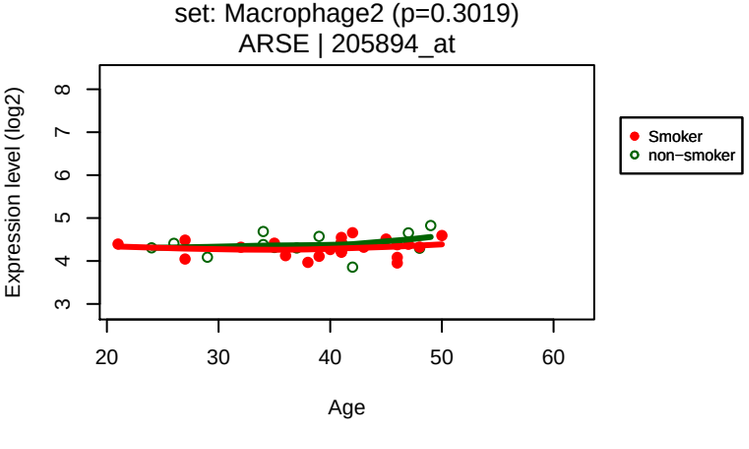

set: SmallAirway2 (p=0.405)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker

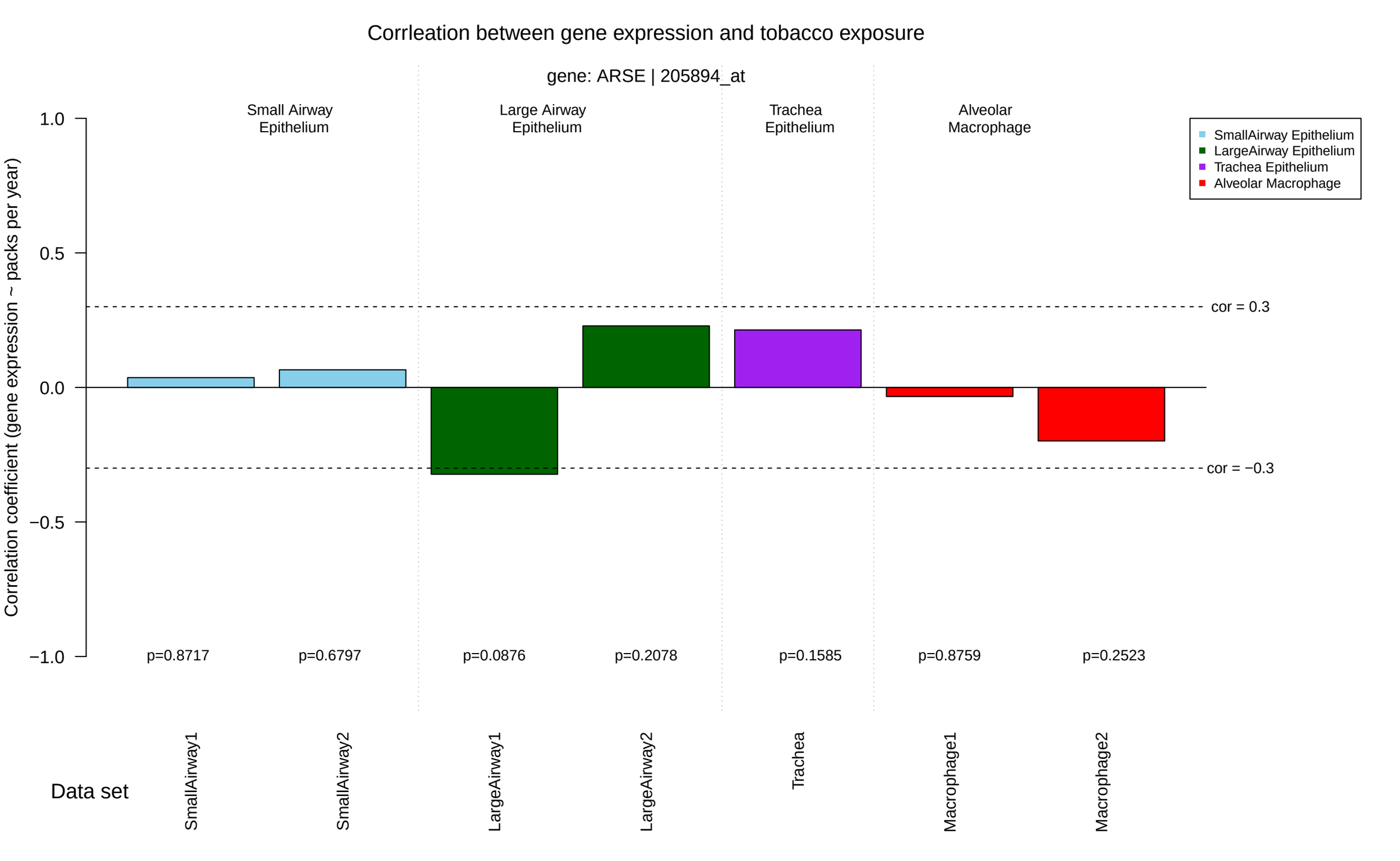

set: LargeAirway1 (p=0.0099)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.345)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3168)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9152)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3019)
ARSE | 205894_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARSE | 205894_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8717
p=0.6797
p=0.0876
p=0.2078
p=0.1585
p=0.8759
p=0.2523
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2